world operates, revealing how organisms obtain the nutrients and other chemicals they require for life from the resources in their environment.

The field of ecological stoichiometry was comprehensively described for the first time by American biologists Robert Sterner and James Elser; in *Ecological Stoichiometry* (2002), they used mathematical models to demonstrate the application at every level, from molecules and cells to individual plants and animals, populations, communities, and ecosystems.

“Individual organisms also show differences in stoichiometry during their life cycles. Young organisms may have different compositions from older ones …”

Robert Sterner and James Elser

Key chemicals

In ecological research, the three main elements examined are carbon (C), nitrogen (N), and phosphorus (P), because each plays a vital role. Carbon is a basic building block of all life and an important part of many chemical processes. Nitrogen is a major constituent of all proteins, while phosphorus is crucial for cell development and storing energy.

An organism’s C:N:P ratio is not necessarily consistent. Plants have a variable ratio: they can adjust the balance of their elements according to their environment. For instance, the proportion of carbon in their chemical makeup can rise on a particularly sunny day because more photosynthesis occurs—the process by which they take carbon dioxide from the air and use the sun’s energy to convert it into the nutrients they require.

Higher up the food chain, animals have largely fixed C:N:P ratios, so they must deploy various mechanisms to deal with any imbalances of chemicals entering the body. If an insect or animal herbivore is getting too much carbon from its plant diet, for instance, it may adjust its digestive enzymes and excrete it, store it as fats, or raise its metabolic rate to burn it